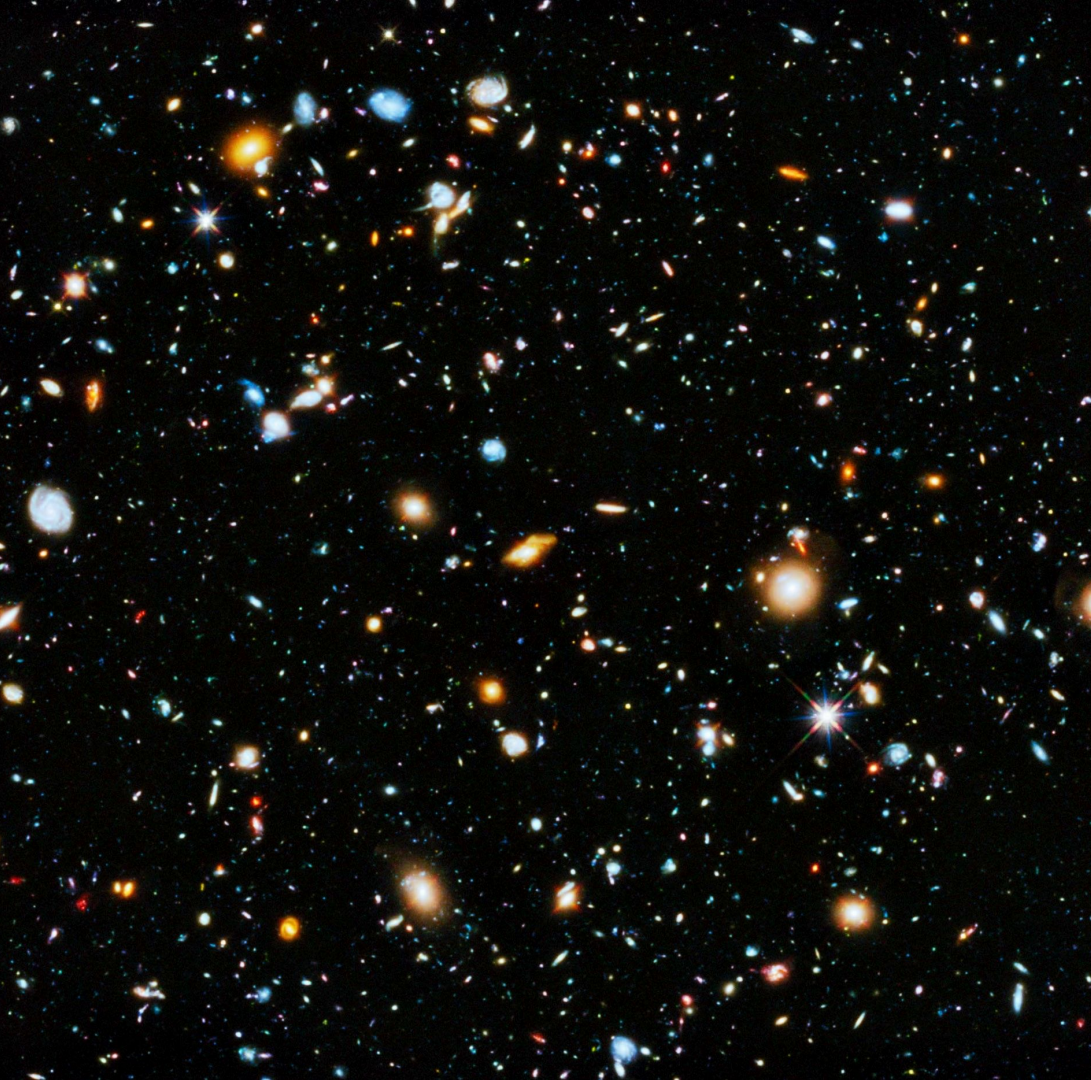


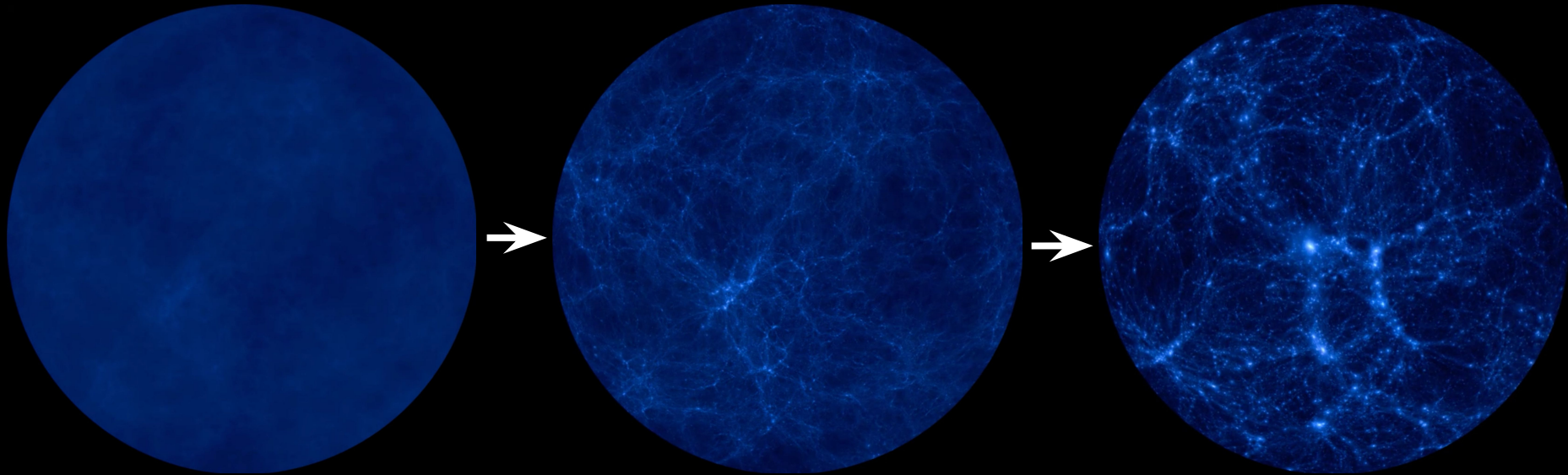


Gravitational Simulations at Cosmic Scales

Lev Kruglyak, Jessica Quaye,
Aditya Karthik Saravanakumar

Large Structure Formation in the Universe

- Start with an initially uniform universe.
- Small perturbations introduce inhomogeneities.
- Energy density and pressure drives spatial expansion ($a(t)$ factor).
- As universe expands, small perturbations form into massive structures.



Physical Setup

- Use Newtonian approximation to general relativity.
- Matter field approximated by N particles interacting gravitationally.

We hope to simulate $N \sim 10^8 - 10^9$ particles.

Maybe: Use Smoothed Particle Hydrodynamics (SPH)

Newton's Law of Gravity

$$F = G \frac{m_1 m_2}{r^2}$$

Poisson Equation

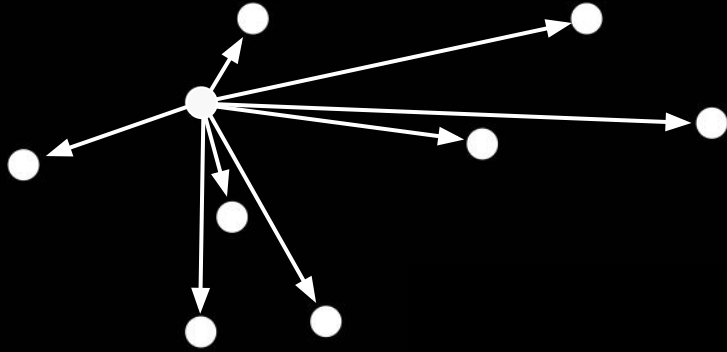
$$\nabla^2 \Phi = 4\pi G \rho$$

Friedmann Equations

$$\left(\frac{\dot{a}(t)}{a(t)} \right)^2 = \frac{8\pi G}{3} \rho(t) - \frac{\kappa c^2}{a^2(t)} + \frac{\Lambda c^2}{3}$$

$$\frac{\ddot{a}(t)}{a(t)} = - \frac{4\pi G}{3} \left(\rho(t) + \frac{3p(t)}{c^2} \right) + \frac{\Lambda c^2}{3}$$

Idea #1: Brute Force Algorithm



Every particle computes the gravitational force from every other particle.

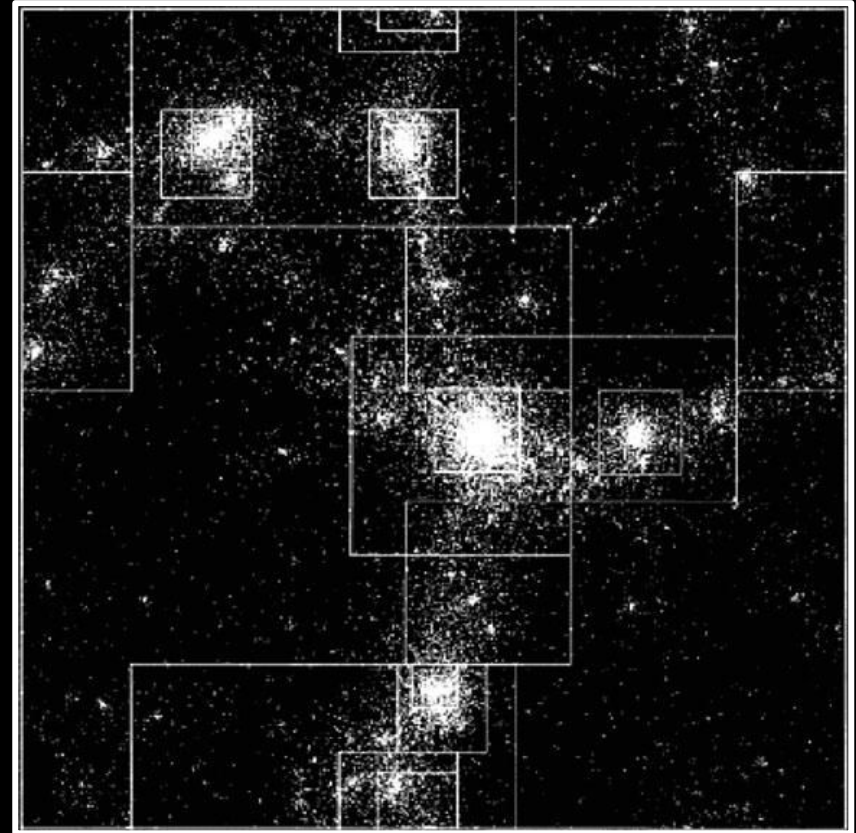
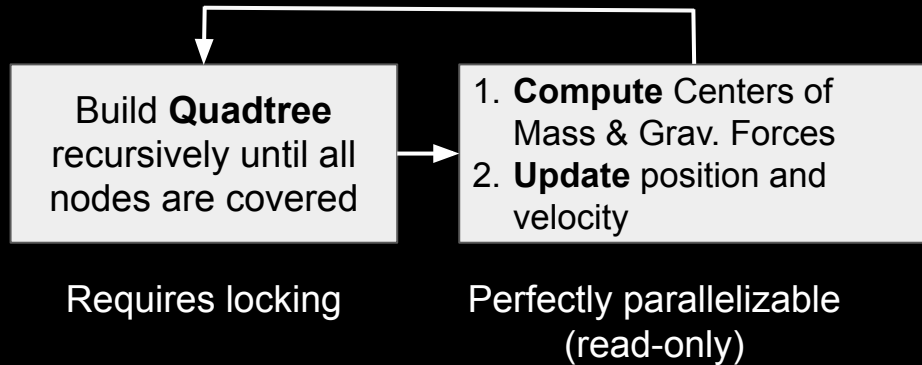
$O(N^2)$ computational complexity which is impractical for large N .

Use Verlet, a symplectic integrator, (more stable than Euler method for energy conservation) to update positions and velocities.

Idea #2: Tree-Based Algorithm (e.g. Barnes-Hut)

$O(N \log N)$ algorithm.

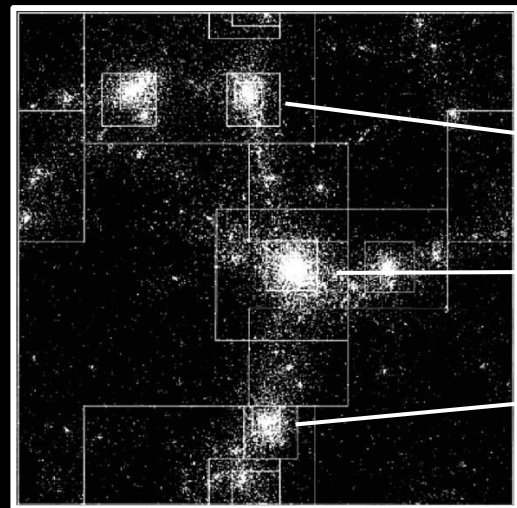
Spatial subdivision structure approximates group of points as a single large body.



Our Plan

To get truly large simulations, we parallelize at two scales.

Variable timesteps avoid synchronization overhead.



Local OpenMP
N-Body Solver

Node 1

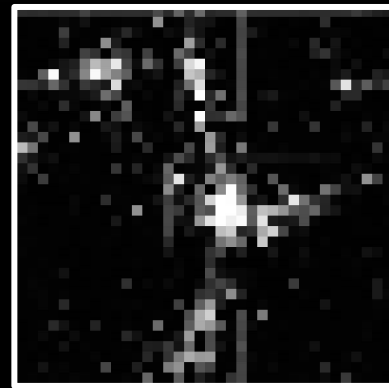
Node 2

Node 3

Each region gets its own node,
does a high precision tree-based
solver at a low timestep

MPI

$$\nabla^2 \Phi = 4\pi G \rho$$



Low-resolution
density mesh

Poisson Solver

Approximate Poisson
solver using FFT

Challenges

- Load balancing across nodes
- Domain decomposition
 - Grid boundaries introducing numerical artifacts
 - Reassigning particles to different nodes
 - Synchronization and data transfer
- Simulation files will be gigantic (10^8 particles means 10 Gb a frame)
 - snapshotting
 - compression
 - separate node responsible for exporting OpenVDB (sparse volumetric data format)

Preliminary Results - Serial Implementation

